

# COURSE SYLLABUS

CS310 : Foundations of Computer Science

## COURSE DESCRIPTION

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>Credit points</b>                        | 3 credits (45 hrs)                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
| <b>Level</b>                                | Undergraduate                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
| <b>Instructor</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Teaching time<br/>Location</b>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Prerequisites</b>                        | MTH 1112                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
| <b>Recommended<br/>background knowledge</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |
| <b>Subject description</b>                  | A broad perspective of computer science concepts intended as preparation for more in-depth coverage in higher-level courses. Topics include machine and assembly language programming, computer system organization and operation, logic circuits, finite-state diagrams and programming language grammar, Boolean algebra, and circuit design considerations.                                                                                                         |     |
| <b>Objectives &amp; Out-come</b>            | Introduction to information technology and computer science. Computer science: definition, theory, and applications. Algorithmic foundations of computer science. Representing algorithms, pseudocode, sequential operations, efficiency, and attributes of algorithms. System Software. Assemblers and assembly language. Operating systems. Introduction to high-level language programming. Computer systems organization. binary numbers, Boolean logic and gates. |     |
| <b>Assessment/ Evaluation</b>               | Attendance/Attitude                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10% |
|                                             | Group presentation                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 20% |
|                                             | Mid-term exam                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 30% |
|                                             | Final exam                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 40% |
| <b>Prescribed Textbook(s)</b>               | [1] Invitation to Computer Science, 7 <sup>th</sup> Edition, G.Michael Schneider, Judith Gersting, 2015.<br>[2] Introduction to Logic Design, 3 <sup>rd</sup> Edition, Alan B. Marcovitz, McGraw-Hill, 2010.                                                                                                                                                                                                                                                           |     |

## COURSE CONTENTS & SCHEDULE

| Class No. | Contents                                                                                                                                                                                                                                             | No. of Hours |      |      | Ref./Resources            | Assignment(s) |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------|------|---------------------------|---------------|
|           |                                                                                                                                                                                                                                                      | Lect.        | Exr. | Prc. |                           |               |
| 1         | <b>Course Introduction and Orientation</b><br><b>Chapter 1: Introduction</b><br>1.1 Introduction<br>1.2 Computer Science: Definition, Theory and Applications<br>1.3 A Brief History of Computing<br>1.4 Some of the Major Advancements in Computing | 3            |      |      | Chapter 1<br>[Textbook 1] | 2             |
| 2         | <b>Chapter 2: Algorithmic Foundations of Computer Science</b><br>2.1 Introduction<br>2.2 Representing Algorithms<br>2.3 Pseudocode<br>2.4 Sequential Operations                                                                                      | 3            |      |      | Chapter 2<br>[Textbook 1] | 2             |
| 3         | 2.5 The Efficiency of Algorithms<br>2.6 Attributes of Algorithms<br>2.7 Measuring Efficiency<br>2.8 Order of Magnitude—Order $n$<br>2.9 Order of Magnitude—Order $n^2$<br>2.10 Analysis of Algorithms                                                | 3            |      |      | Chapter 3<br>[Textbook 1] | 3             |
| 4         | Exercise and in-class assignment                                                                                                                                                                                                                     |              | 3    |      |                           | Assignment 1  |
| 5         | <b>Chapter 3: System Software and Virtual Machines</b><br>3.1 Introduction<br>3.2 System Software<br>3.3 The Virtual Machine<br>3.4 Types of System Software<br>3.5 Assemblers and Assembly Language<br>3.6 Operating Systems                        | 3            |      |      | Chapter 6<br>[Textbook 1] | 2             |
| 6         | <b>Chapter 4: High-Level Language Programming</b><br>4.1 The Language Progression<br>4.2 A Family of Languages                                                                                                                                       | 3            |      |      | Chapter 9<br>[Textbook 1] | 2             |

|    |                                                                                                                                                                                                                                                                                                                                               |   |   |  |                               |                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|-------------------------------|---------------------|
|    | 4.3 Feature Analysis<br>4.4 Meeting Expectations                                                                                                                                                                                                                                                                                              |   |   |  |                               |                     |
| 7  | 4.5 Software Engineering<br>4.6 The Software Development Life Cycle<br>4.7 Procedural Languages (Fortran, COBOL, C/C++, Ada, Java, Python, C# and .NET)<br>4.8 Special-purpose Languages (SQL, HTML)                                                                                                                                          | 3 |   |  | Chapter 9, 10<br>[Textbook 1] | 1                   |
| 8  | Assignment 1 submission<br>Practice and in-class assignment                                                                                                                                                                                                                                                                                   |   | 3 |  |                               | Assignment 2        |
| 9  | <b>Mid-term exam</b><br>Exam solution                                                                                                                                                                                                                                                                                                         |   | 3 |  |                               |                     |
| 10 | <b>Chapter 5: The Building Blocks: Binary Numbers, Boolean Logic, and Gates</b><br>5.1 Introduction<br>5.2 The Binary Numbering System<br>5.2.1 Binary Representation of Numeric and Textual Information<br>5.2.2 Binary Representation of Sound and Images<br>5.2.3 The Reliability of Binary Representation<br>5.2.4 Binary Storage Devices | 3 |   |  | Chapter 4<br>[Textbook 1]     | 2                   |
| 11 | 5.3 Boolean Logic and Gates<br>5.3.1 Boolean Logic<br>5.3.2 Gates<br>5.4 Combinational and Sequential Circuits                                                                                                                                                                                                                                | 3 |   |  |                               | 2                   |
| 12 | <b>Chapter 6: Computer Systems Organization</b><br>6.1 Introduction<br>6.2 The Components of a Computer System<br>6.2.1 Memory and Cache<br>6.2.2 Input/Output and Mass Storage<br>6.2.3 The Arithmetic/Logic Unit<br>6.2.4 The Control Unit                                                                                                  | 3 |   |  | Chapter 5<br>[Textbook 1]     | 2                   |
| 13 | 6.3 Putting All the Pieces Together—the Von Neumann Architecture                                                                                                                                                                                                                                                                              | 3 |   |  |                               | Assignment 3: Group |

|    |                                    |  |   |  |  |              |
|----|------------------------------------|--|---|--|--|--------------|
|    | 6.4 Non–Von Neumann Architectures  |  |   |  |  | presentation |
| 14 | <b>Group presentation</b>          |  | 3 |  |  |              |
| 15 | <b>Final exam</b><br>Exam solution |  | 3 |  |  |              |

*Notes:*

- *Abbreviation: Lect. (lecture), Exr. (Exercise), Prc. (Practise).*
- *Exercises may include assignment, reports, student's presentation, homework, class exercises... for each class sessions.*
- *Practical mostly refer to Lab-work or outside practice such as field trip.*
- *Assignments may include assignments, practical work, reports, exercises ...for each class sessions*

#### **Reference Literature:**

- [1] Invitation to Computer Science, 7<sup>th</sup> Edition, G.Michael Schneider, Judith Gersting, 2015.
- [2] Introduction to Logic Design, 3<sup>rd</sup> Edition, Alan B. Marcovitz, McGraw-Hill, 2010.